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1: Ethical Business Plan

1.A. Company Name

The name of our company is Game Sector. We chose this name because our goal is to create a fair, skill-based multiplayer game where competition is driven by strategy and improvement rather than spending money, reflecting our focus on integrity and balanced gameplay.

1.B. Long Term Vision Statement

1.B.1 Goals

Our main goal is to create a strategic game that is fun, fair, and accessible to all players. We want to show that games can be engaging without using pay-to-win systems, manipulative design, or unnecessary data collection. Over time, we hope to influence the gaming industry by proving that ethical game design can still lead to a successful and enjoyable product.

1.B.2 Idea Origination

Our idea came from our own experiences as gamers and from class discussions about ethics in technology. We noticed how many games rely on addictive mechanics or unfair monetization, and we wanted to create something that avoids those problems. Additionally, we want to make ourselves transparent and let players know that their information when playing our game is safe and will not be misused for the company's benefit. This pushed us to build a strategic game that focuses on skill, fairness, and player well-being.

1.B.3 Purpose/Values/Mission

Our mission is to create an accessible and ethical gaming experience that also prioritizes the player's well being, data privacy, and fair play. Game Sector exists to design a game that is fun, engaging, and safe to the player. Our purpose is to push the gaming industry toward healthier design practices by avoiding manipulative monetization, protecting user data/info, and ensuring inclusive, enjoyable gameplay for all players. We value fairness, transparency, and respect for the player.

1.B.4 Key Questions

How can we design a strategic game that stays fair without using pay-to-win mechanics?

How do we protect the player data while still offering a good gameplay experience?

What design choices would help keep players safe from addictive or manipulative features?

1.C. Strategy with Ethical Impacts and Ethical Safeguards

1.C.1 OKR 1

1.C.1.1 Objective and Key Result

As an example of the ethical commitments described above, in-game cosmetic purchases directly impact player's experiences, one of the stakeholders of our company, along with the investors in Game Sector, another key stakeholder. The handling of player's payment information must be done with utmost attention to security, and the items available for purchase must be as appealing as possible to players to fulfill our fiduciary duty to shareholders. These items must not offer a gameplay advantage, as that would punish players who decide not to purchase them, and their allocation mustn't be the result of a randomized reward, as that would function essentially as gambling.

1.C.1.2 Metric(s) with Experiment(s)

The metric we will use to evaluate the efficacy of our in-game purchases will be the sales of our in-game products as compared with those of games with similar player bases that do use mechanics like in-game currency and lootboxes. It is our expectation that sales of these items will be as high or better than these other games if they are designed with enough visual flair and appeal.

1.C.1.3 Ethical Impact(s)/Issue(s)

Any online service that involves payment processing runs the risk of being breached by online hackers. Sony suffered such an attack and gave sensitive information of thousands of customers to the hackers [1]. Game Sector will take this responsibility seriously and will employ strict security measures to keep our customer's data private.

1.C.1.4 Ethical Safeguards

We will hire experts in the field of cyber security along with researchers involved in the science of addiction to ensure our games are designed ethically. The researcher B.F. Skinner was a psychologist who believed that controlling an organism's surrounding stimuli was the key to controlling its behavior [2]. He used experiments with mice in which they were split into 2 groups, one of whom received a consistent reward when pressing a button, the other which received a reward at random when pressing a button. The second group became obsessed with the button, and this forms the basis of many unethical game's design. We hope to avoid this scenario by employing experts in the field of psychology.

1.C.2 OKR 2

1.C.2.1 Objective and Key Result

Objective

An objective that my team's company is trying to achieve is to develop strong data-privacy practices that minimize data collection and keep players informed about how their information is used.

Key Result & Stakeholders

A key result that supports this objective is achieving 90% player comprehension on the data-transparency pop up during usability testing. This key result focuses on making sure players clearly understand what information the game collects and why. According to the research done by the Pew Research Center, "Many Americans have little to no understanding of what companies are doing with the data that is collected about them" and their survey found that "just 9% of adults say they always read a company's privacy policy before agreeing to the terms and conditions...More than a third of adults (36%) say they never read a privacy policy before agreeing to it" [1], which reinforces why our transparency pop-up must be simple and easy for players to understand.

The primary stakeholders are the players themselves, who interact with the transparency pop-up during account creation or when new features require additional data. Their ability to make informed decisions depends on how clearly information is presented. Parents and guardians are also important stakeholders, especially for younger players who may not fully understand privacy concepts. They rely on clear explanations to determine whether our game is safe for their children.

Our customer base is diverse, including teenagers, college students, and adults who enjoy competitive strategy games. Players come from a wide range of racial and gender backgrounds, and income levels vary from lower-income students to working adults. Many players are interested in competitive gaming, skill-based progression systems, and being a part of Esports/competitive teams. And because our game avoids pay-to-win mechanics, it attracts players who value fairness and transparency, making clear communication especially important for maintaining trust.

Additional stakeholders include the company itself, which must design and test the transparency interface; cloud service providers that store user data; and platform companies such as the Apple App Store and Google Play, which enforce their own privacy and data-handling requirements. Government and regulatory bodies like COPPA, GDPR, and CCPA also shape what information can be collected and how it must be disclosed. These groups are connected through shared responsibilities: players depend on Game Sector to protect their information, parents depend on both the company and regulators to ensure safe use for minors, and the company depends on cloud providers and platform policies to remain compliant. Achieving this key result strengthens trust across all stakeholders by demonstrating that players genuinely understand how their data is handled.

1.C.2.2 Metric(s) with Experiment(s)

Success for this OKR will be measured by how well players understand the data-transparency pop-up after seeing it. The main metric we will use is the player comprehension rate, which tells us the percentage of users who correctly understand what data the game collects and why. To measure this, we will run a usability study where players go through a short account-creation flow and then answer a simple five question quiz about the pop-up. For example, we will recruit around 150 players from our target audience and ask questions about what data is collected, how it is stored, and whether players can opt out of certain types of data. If at least 90% of participants answer four out of the five questions correctly (80%), then we will consider the Key Result achieved. If the score is lower, we will revise the pop-up to make it clearer and run the test again.

Here is the actual quiz we will use:

Q1. What types of data does Game Sector collect?

- A. Email and gameplay statistics
- B. Your phone contacts
- C. Your photos
- D. Your text messages

Correct Answer: A

Q2. Why does the game collect gameplay statistics?

- A. To sell data to advertisers
- B. To improve fairness and match players by skill
- C. To track your location
- D. For unrelated marketing purposes

Correct Answer: B

Q3. Where is your data stored?

- A. On unsecured public servers
- B. On Game Sector's secure cloud servers
- C. On every player's device
- D. On social media platforms

Correct Answer: B

Q4. Can players opt out of certain data collection?

- A. Yes, for non essential data
- B. No, players have no control
- C. Only if they pay for premium
- D. Only if they delete their account

Correct Answer: A

Q5. When does Game Sector notify players about data changes?

- A. Never
- B. Only after complaints
- C. Whenever new features require additional data
- D. Only once per year

Correct Answer: C

A second metric we will use is the task success rate, which focuses on whether players can simply explain the data-collection practices in their own words. This helps us see if players truly understand the information rather than just selecting the right quiz answers. To measure this, we will run a smaller think-aloud study with about 50 participants. After viewing the pop-up, each person will be asked to describe what data the game collects and why it is needed. Their responses will be reviewed and scored as either fully correct, partially correct, or incorrect. Our goal is for at least 85% of the participants to give a fully correct explanation. A fully correct

response must simply explain both what data is collected and why. If many players struggle, that will show us that the pop-up still needs improvement.

Together, these two experiments help us measure both accuracy and real understanding. The quiz gives us a clear percentage score and the think-aloud interviews show whether players can actually explain the information. Using both methods helps us make sure our data-privacy practices are clear, transparent, and easy for players to understand.

1.C.2.3 Ethical Impact(s)/Issue(s)

Pursuing this OKR brings several ethical implications and potential issues. Since our goal is to minimize data collection and keep players informed about how their information is used, we have to consider what could happen if players misunderstand the transparency pop-up or if the information is not communicated clearly enough. Even with good intentions, unclear data practices can lead to players unknowingly agreeing to something they did not fully understand, which raises serious ethical concerns around informed consent and privacy.

One of the main concerns is privacy. If the transparency pop-up is confusing or too long, players might skip through it without realizing what data the game collects. This could lead to situations where players believe the game only collects basic account information, when in reality it also collects gameplay statistics or device information. A real-world case that highlights this risk is the EPIC v. Presidential Advisory Commission on Election Integrity lawsuit. In that case, the Commission attempted to gather extensive voter data without proper notice or transparency, and the complaint stated that the Commission “deprived EPIC’s members of their liberty interest in avoiding the disclosure of personal matters...without providing notice” [2]. This example shows how failing to clearly communicate data practices can violate people’s privacy and damage trust, even when the organization claims to be acting responsibly.

Another concern is the possibility of players giving uninformed consent. For example, a younger player might assume the pop-up is just a normal game message and click “I agree” without reading anything. A parent might also misunderstand the pop-up and believe the game collects less data than it actually does. These scenarios can lead to players feeling misled or exposed if they later discover the full extent of the data collection. Even if the data collected is minimal, the ethical issue remains the same: players deserve to understand what they are agreeing to.

Stake Holder	Financial	Privacy	Conflicting Interest	Violation of Rights
Players	Low	High	Mid	High

Parents/Guardians	Low	High	Mid	Mid
Game Sector (Company)	Mid	Mid	High	Mid
Cloud Service Providers	Low	Mid	Low	Low
App Stores (Apple/Google)	Low	Mid	Low	Low
Regulators (COPPA, GDPR, CCPA)	Low	Low	Low	Low

From the Ethical Impact Risk Table shown above, we can see the different risks each stakeholder faces. For players, the financial risk is low because the game does not charge players based on data collection. However, the privacy risk is high. If the pop-up is unclear, players may unknowingly share information they did not intend to. Conflicting interest risk is mid because players want transparency and control, while the company may want to keep the onboarding process simple. Violation of rights risk is high because unclear consent directly affects a player's right to understand and control their personal information.

For parents and guardians, financial risk is also low, but privacy risk is high. Parents are responsible for protecting their child's information, and unclear data practices make that difficult. Conflicting interest risk is mid (medium) because parents want full clarity, while the company may prioritize ease of use. Violation of rights risk is mid because unclear consent can interfere with a parent's ability to make informed decisions for their child.

For Game Sector, financial risk is mid. If players feel misled or confused, the company could face complaints, negative reviews, or even regulatory penalties. Privacy risk is mid because unclear communication can harm the company's reputation. Conflicting interest risk is high since the company must balance transparency with usability, and these goals can conflict. Violation of rights risks is mid (medium) because if players are not properly informed, the company could unintentionally violate privacy expectations or legal requirements.

Cloud service providers face low financial risk and mid privacy risk. They store the data, so any breach or mishandling could affect players, but they do not control what data is collected. Conflicting interest and violation of rights risks are low because their role is mainly technical and governed by contracts.

App stores like Apple and Google face low financial risk and mid privacy risk. They require apps to follow strict data-handling rules, and unclear data practices could violate store policies. Conflicting interest and violation of rights risks are low because they enforce compliance but do not collect the game's data themselves.

Regulators face low risk across all categories. Their role is to enforce privacy laws, not collect or store data. Their main concern is whether the company follows the rules, and they are not directly harmed by unclear data practices.

1.C.2.4 Ethical Safeguards

To reduce the ethical issues described earlier, we will use safeguards that help players clearly understand what data is being collected and why. Since our OKR is about minimizing data collection and keeping players informed, the safeguards need to be simple and easy for anyone to follow. This is important because unclear or overly long explanations can make it harder for people to understand what they are agreeing to. The Federal Trade Commission (FTC) emphasizes this in its report on mobile privacy, stating that "...consumers are unaware that many of these practices are taking place, they do not look for options providing them with control over such practices...Participants noted that when disclosures are made, consumers often do not understand them" [3]. This supports our decision to keep the pop-up short and easy to understand so players can make informed choices.

To prevent confusion and uninformed consent, we will use a short, plain-language pop-up during onboarding. It will include three simple bullet points: what data we collect, why we collect it, and what players can opt out of. A small "Learn More" button will be available for players who want extra detail. A communication-focused peer or a student with UI experience will help review the wording. We will measure effectiveness through the comprehension quiz—if fewer than 90% answer correctly, we will revise and retest.

To make the message even clearer, we will rewrite all data explanations in everyday language. Instead of using terms like "gameplay telemetry", we will say "information about how to play." This helps younger players and parents understand what the pop-up is talking about. We will test this wording with a small group of classmates or volunteers to confirm it was easy to understand. We will use the think-aloud study to measure success. If fewer than 85% can explain the data practices in their own words, we will update the language.

To make sure players are not agreeing to something they don't understand, we will add a simple confirmation step. After reading the pop-up, players will answer one yes or no question like "Do you understand why we collect gameplay information?" A peer with basic UI

experience will help integrate this step smoothly. We will track how many players select “No” and if more than 10% do, we will revise the explanation.

1.C.3 OKR 3

1.C.3.1 Objective and Key Result

The objective will be to launch a successful beta of the strategy game that protects user data, and achieves high satisfaction among a diverse group of players. The goals we have for the beta is reaching 10,000 active beta users within the first three months, achieving a 90% or higher player satisfaction rating through feedback surveys, maintaining a balanced gameplay with no pay to win advantages, and ensure full compliance with data privacy regulations and zero data misuse incidents.

The primary stakeholders in this milestone are the players who serve as the main customer base. These players will primarily be teenagers and young adults from diverse backgrounds, including different income levels, races, and genders. The game is designed to be affordable and accessible, making it inclusive for lower-income players, while also ensuring fair gameplay that does not favor any specific group.

Developers within Game Sector are also key stakeholders, as they are responsible for designing, maintaining, and improving the game. They are affected throughout the entire process. Especially during the beta phase, where they must respond to user feedback, fix issues, and ensure that ethical standards such as fairness and transparency are upheld. Platform providers, such as digital game distribution services, are stakeholders because they host and distribute the game. They are affected at launch and beyond, as they require the game to meet their policies and benefit from offering a trustworthy product. Government and regulatory bodies are also involved, as they ensure compliance with data privacy and consumer protection laws. These organizations influence how the company collects, stores, and manages user data. Additionally, parents and guardians may be stakeholders for younger players as they are concerned with safety, privacy, and healthy gameplay habits. The broader gaming community, including content creators and streamers, also plays a role by promoting, reviewing, and influencing public perception of the game.

All stakeholders are interconnected. Players will provide feedback that developers use to improve the game, while platforms enable access to users. Governments enforce regulations that guide ethical practices, and the gaming community helps shape the game’s reputation. Together these relationships ensure that the company remains accountable while delivering a fair, transparent, and enjoyable gaming experience.

1.C.3.2 Metric(s) with Experiment(s)

To measure the success of our OKR we will define specific, quantifiable metrics and design experiments to evaluate each one. These metrics will allow us to clearly determine whether our goals around fairness, player satisfaction and ethical design are being achieved. The first metric will be a survey with 5 questions that the players will need to answer, the scoring will be from 1 to 5, 1 scoring 0%, 2 scoring 25%, 3 scoring 50%, 4 scoring 75%, and 5 scoring 100%, with our target being 90% or more. The five questions are the, the game felt fair and balanced, I did not feel pressured to spend money to compete, the game was enjoyable and engaging, I trust how my data is being handled, and I would recommend this game to others.

The second metric will be measuring how fair the game is by analyzing win rate distribution across players and strategies. The measurement to determine if it is fair will be the win rate being between 40% and 60%, with the target being no strategy being over a 60% win rate or below a 40% win rate.

The third metric will be measuring whether players believe the game is free from pay to win advantages. The survey question will be, “Do You believe spending money gives players a competitive advantage?”, with the metric being players answering No, and the target being 95% or more.

The fourth metric is measuring how confident players feel about how their data is handled. The survey question will be, “I feel confident that my personal data is secure and not misused.” The scoring will be the same as the first metric, with the target being 90% or more. The fifth metric will measure how many players continue to play the game over time. The formula will be a 7 day retention rate with the formula for it, $(\text{Number of players who return after 7 days} / \text{total players}) * 100$. The target for this formula is 60% or more.

1.C.3.3 Ethical Impact(s)/Issue(s)

Achieving our OKR introduces several ethical risks despite our intentions. Even with safeguards, issues related to fairness, privacy, and stakeholder interests may arise. The first issue is Data privacy and security risks. Even though Game Sector minimizes data collection, user data (such as login credentials and gameplay statistics) is still stored on servers. A scenario could be, a data breach occurs during the beta phase, exposing user account information. Even if sensitive data is limited, players may lose trust in the platform. Research shows that even small scale breaches can significantly impact user trust and company reputation [1]. The second issue is algorithmic fairness and bias. The matchmaking and balancing system may unintentionally favor certain players or strategies. A scenario could be, A specific strategy becomes dominant due to a

flaw in balancing, giving some players an unfair advantage and violating the “fair play” goal. Algorithmic systems can unintentionally create bias if not continuously monitored and tested [2]. The third issue is player well being and addiction, even without manipulative monetization, games can still encourage excessive play. A scenario could be, players spend long hours playing the game due to competitive ranking systems, negatively impacting their academic or personal lives. Studies show that game design elements can influence compulsive usage patterns even without explicit monetization [3]. The fourth issue is transparency and misleading design. If players do not fully understand how systems work, trust can be reduced. A scenario could be, players believe the game is unfair due to a lack of transparency in matchmaking, even if the system is balanced. The fifth issue is conflicting business interests. There may be pressure to introduce monetization features that conflict with ethical goals. A scenario could be, investors push for revenue growth, encouraging the addition of features that may indirectly create advantages for paying users. Companies often face tension between profit goals and ethical responsibilities [4]. Here is an expected ethical impact risk table

Stakeholder	Financial	Privacy	Conflicting Interests	Violation of Rights
Players	Low	Mid	Mid	Low
Developers	Mid	Low	High	Low
Platform Providers	Low	Low	Mid	Low
Government/Regulators	None	Low	Low	None
Parents/Guardians	Low	Mid	Low	Low
Gaming Community	None	None	Mid	None

The players' finances are at a low because there is no pay to win system which reduces financial harm. The privacy is at mid because some data is still collected and could be breached. The conflicting interest is at mid because company goals may conflict with fairness. The violation of Rights is at low because minimal risk if policies are followed.

The developers' finances are at mid because the job pressure is tied to product success. The privacy is at low because there is limited personal risk. The conflicting interests are at high because of pressure between ethical design vs profit goals. The violation of rights is at low because there is no direct rights concerns.

The platform providers' finances are at low because there is revenue from hosting but low direct risk. The privacy is at low because there is limited involvement in data handling. The conflicting interest is at mid because it may prioritize profitable games over ethical ones. The violation of rights is at low because there is minimal direct impact.

The government/regulators' finances are at none because there is no direct financial stake. The privacy is at low because they only have an oversight role. The conflicting interest is at low because they are primarily focusing on enforcement. The violation of rights is at none because their role is to protect rights.

The parents/guardians' finances are at low because there is minimal spending expected. The privacy is at mid because there is concern over children's data. The conflicting interest is at a low because they want safety over engagement. The violation of rights is at a low because there are risks if the child's data is mishandled.

The gaming community's finances are at a none because they have no direct dependency. Their privacy is at none because it is not directly affected. Their conflicting interest is at mid because they may favor hype over ethical critique. The violation of rights is at none because there is no direct rights concerns.

1.C.3.4 Ethical Safeguards

To address the ethical risks identified in our OKR such as data privacy concerns, fairness issues, player well being, and conflicting business interests, Game Sector will implement a set of structured safeguards. These safeguards are designed to prevent, reduce, and continuously monitor ethical harms while maintaining a high quality player experience.

The first one being Data privacy and security safeguard, while Game Sector will implement a data minimization and encryption framework, where only essential user data is collected and stored. All data will be encrypted both in transit and at rest. Players will also have access to a clear privacy dashboard where they can view, download, or delete their data at any time. The people involved will be software engineers, cybersecurity experts, and legal/compliance experts. Experts can be recruited through industry hiring platforms or external security consulting firms. It will be implemented by using industry standard encryption protocols, develop a user facing privacy dashboard, and conduct regular security audits and

penetration testing. We will measure effectiveness by the number of data breaches, user trust score from surveys, percentage of users engaging with privacy controls.

The second one is fair gameplay and anti-bias safeguard. To ensure fairness, the game will use a continuous balancing system supported by analytics that monitor win rates, strategy usage, and matchmaking outcomes. Any imbalance will trigger automatic review and updates. Ethical game design research emphasizes the importance of balancing gameplay to promote fairness and collaboration among players [5]. The people involved will be game designers and developers, data analysts, and external testers. These stakeholders ensure that different playstyles and demographics are considered. It will be implemented by monitoring gameplay data in real time, run regular balance patches, and include player feedback tools. It will be measured by the win rate distribution staying within 40%-60%, reduction in player complaints about fairness, and survey based fairness score.

The third safeguard is Player well being. Game Sector will integrate healthy gameplay features, such as optional playtime reminders, session limits, and break suggestions to reduce excessive use. Prior studies in ethical game design highlight that considering player well being enhances both satisfaction and long term engagement [5]. The people involved will be UX designers, behavioral experts, and player feedback groups. Experts can be consulted through academic partnerships or research institutions. This will be implemented by adding optional notifications after extended play sessions, avoid addictive mechanics, and provide gameplay summaries to encourage reflections. We will measure the effectiveness by the average session length trends, retention rate balanced with healthy usage, and player feedback on well being features.

The fourth safeguard is transparency and communication. The company will adopt a transparency first approach, clearly explaining how matchmaking, monetization, and data usage work. Transparency is a key principle in ethical game design to build trust and fairness among players [5]. The people involved will be developers, UX teams, and legal advisors. This will be implemented by providing simple explanations during onboarding, publishing transparency reports, and including in-game tooltips explaining systems. We will measure effectiveness with player trust scores with it being 90% or over, reduction in confusion related complaints, and engagement with transparency features.